

HW # 20

M. Abdullah Siddiqui 23I-0617 CS-F

Problem 1:

$$M_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1$$

$$C_{11} = (-1)^{1+1}(-1) = -1$$

$$M_{12} = -5$$

$$C_{12} = (-1)(-5) = 5$$

$$M_{13} = 4$$

$$C_{13} = 1(4) = 4$$

$$M_{21} = 2$$

$$C_{21} = -1(2) = -2$$

$$M_{22} = -4$$

$$C_{22} = 1(-4) = -4$$

$$M_{23} = -8$$

$$C_{23} = -1(-8) = 8$$

$$M_{31} = 5$$

$$C_{31} = 1(5) = 5$$

$$M_{32} = -3$$

$$C_{32} = -1(-3) = 3$$

$$M_{33} = -6$$

$$C_{33} = 1(-6) = -6$$

Problem 2:

$$|A| = \begin{vmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{vmatrix}$$

$$|A| = 0 \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} + 1 \begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix}$$

$$= 0 - 2(-5) + 1(4)$$

$$= 14$$

Problem 3:

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{ccc} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{array} \quad \begin{array}{ccc} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{array}$$

$+4 \times 0 - 0 \times 6 = -2$
 $0 + 16 + 0 = 16$
 $|A| = 16 - 2 = 14$

Problem 4:

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$|A| = 0 + 16 + 0 - (-4) - 6 - 0$$

$$= 16 + 4 - 6$$

$$|A| = 14$$

Problem 5:

$$|A| = \begin{vmatrix} \oplus & \ominus & \oplus & \ominus \\ 1 & 2 & 3 & 0 \\ -1 & -1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{vmatrix}$$

expanding
from Col 3

$$= 3 \begin{vmatrix} -1 & -1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}$$

expanding
from col 1

$$= 3 \left(-1(-16) + 3(-7) \right)$$

$$= 3(16 - 21)$$

$$|A| = -15$$